



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

Javascript Insights – Java Script Execution for Selenium

FEB, 2025



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What is Javascript Executor?



What is Javascript Executor?

What is JavaScript Executor?

- JavaScript Executor is an Interface that helps to execute JavaScript through Selenium Web driver.
- JavaScript Executor provides two methods "execute script" & "executeAsyncScript" to run JavaScript on the selected window or current page.



Importance of using Javascript in Selenium



Importance of using Javascript in Selenium

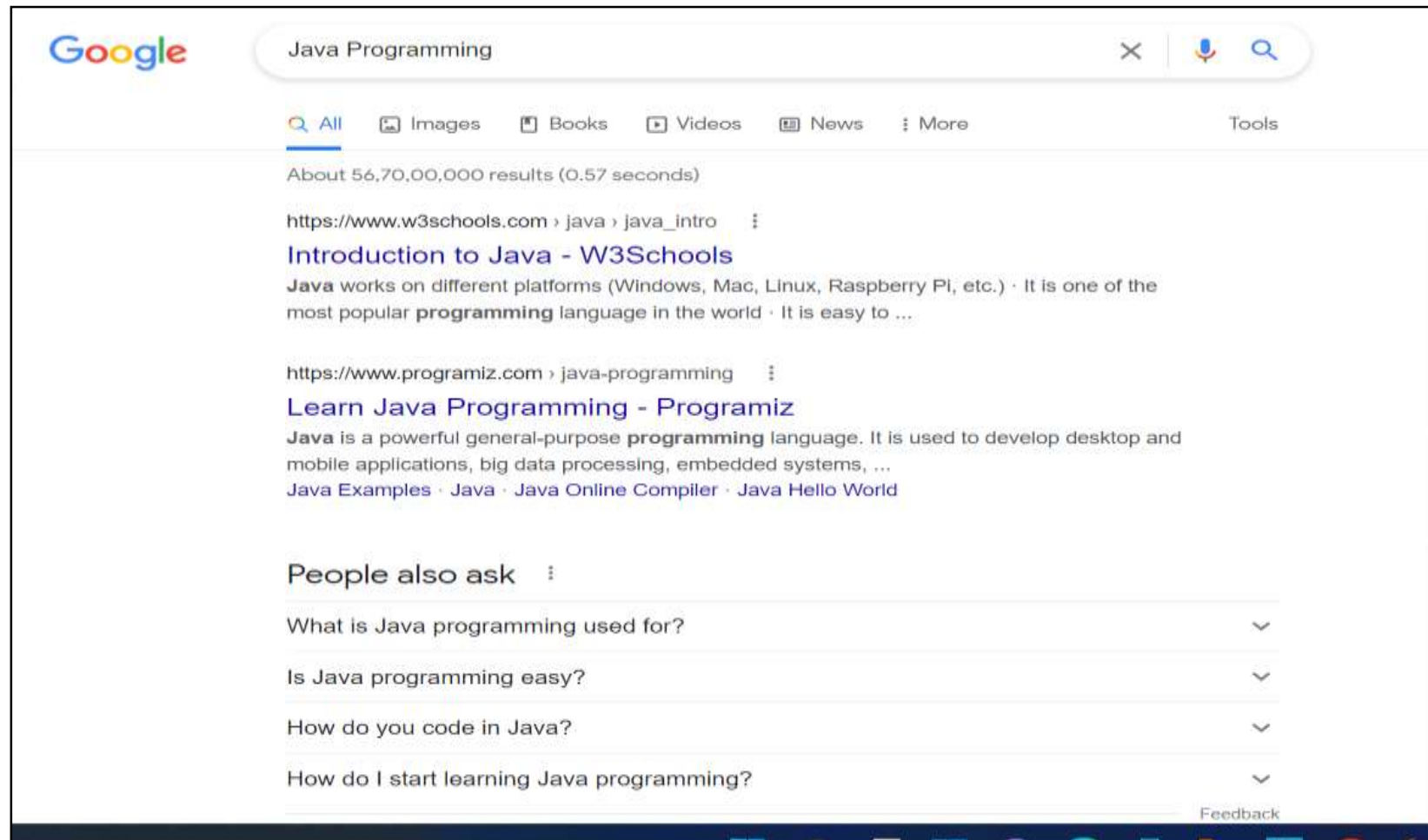
Importance of using Java script in Selenium

- In Selenium Web driver, locators like XPath, CSS, etc. are used to identify and perform operations on a web page
- In case, these locators do not work you can use JavaScript Executor. JavaScript Executor is used to perform a desired operation on a web element
- It provides advantages over Find Element method while handling tricky XPath as well as finding element which are sometimes hidden
- You can directly interact and alter elements on the page

Program 1

```
public class JSDemo2 {  
    public static void main(String args[])  
    {  
        WebDriver driver= new ChromeDriver();  
        driver.get("https://www.google.com");  
  
        JavascriptExecutor js = (JavascriptExecutor)driver;  
        js.executeScript("document.getElementsByName('q')[0].value='Java Programming'");  
        js.executeScript("document.getElementsByName('btnK')[0].click();");  
  
    }  
}
```


Output



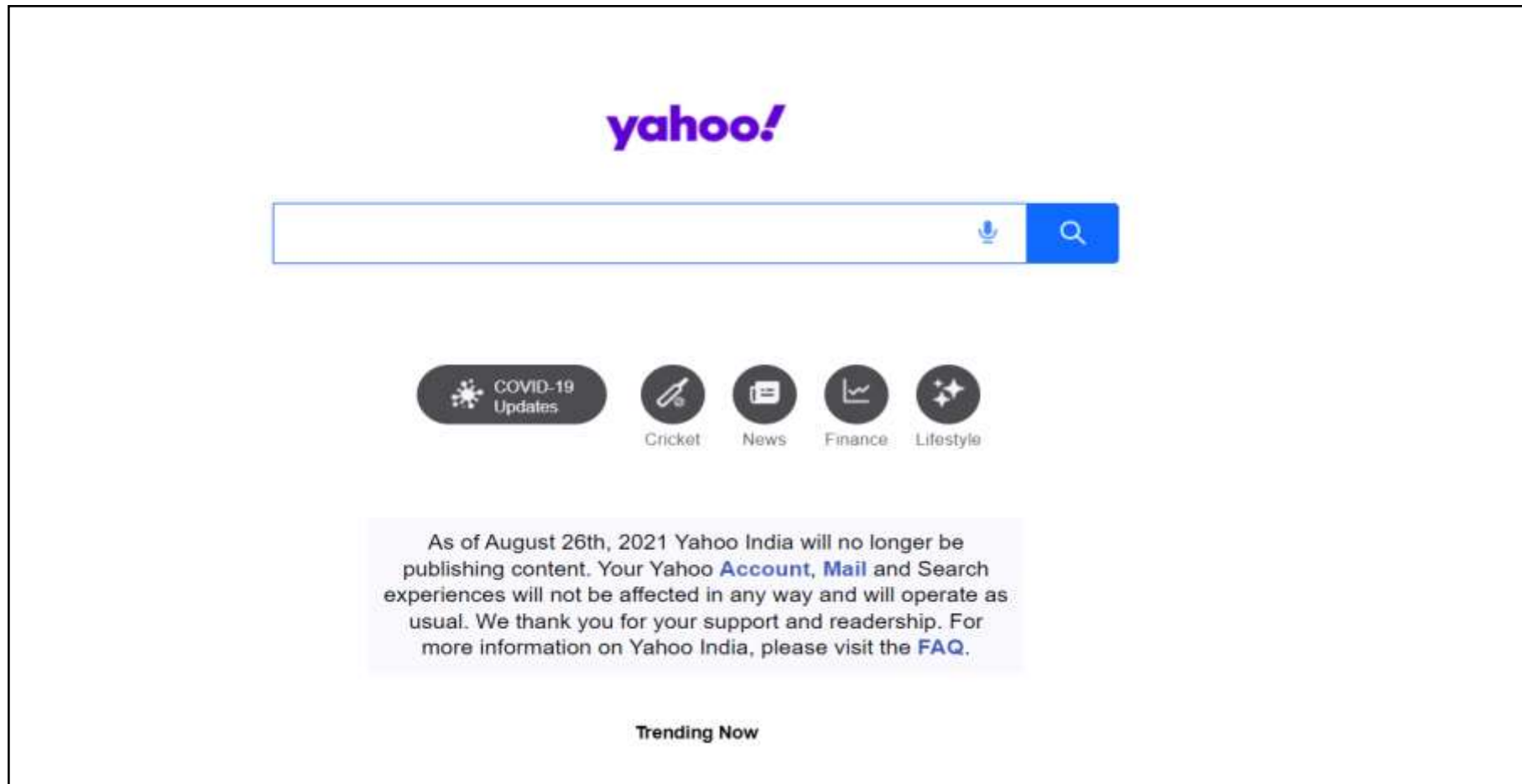
Program 2

```
public class JSDemo3 {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        WebDriver driver= new ChromeDriver();  
        driver.get("https://www.google.com");  
        driver.manage().window().maximize();  
        JavascriptExecutor js = (JavascriptExecutor)driver;  
        //to get the domain  
        String domain = js.executeScript("return document.domain;").toString();  
        System.out.println(domain);  
        //to get the URL  
        String url = js.executeScript("return document.URL;").toString();  
    }  
}
```

Program 2 (Contd.)

```
System.out.println(url);  
//Vertical scroll  
js.executeScript("window.scrollTo(0,100)");  
//scroll to the bottom of the page  
js.executeScript("window.scrollTo(0,document.body.scrollHeight)");  
    //Navigate to new Page  
    js.executeScript("window.location='https://www.yahoo.com'");  
    //to refresh browser window using Javascript  
    js.executeScript("history.go(0)");  
//to generate Alert Pop window in selenium  
js.executeScript("alert('hello world');");  
driver.switchTo().alert().accept();  
}  
}
```

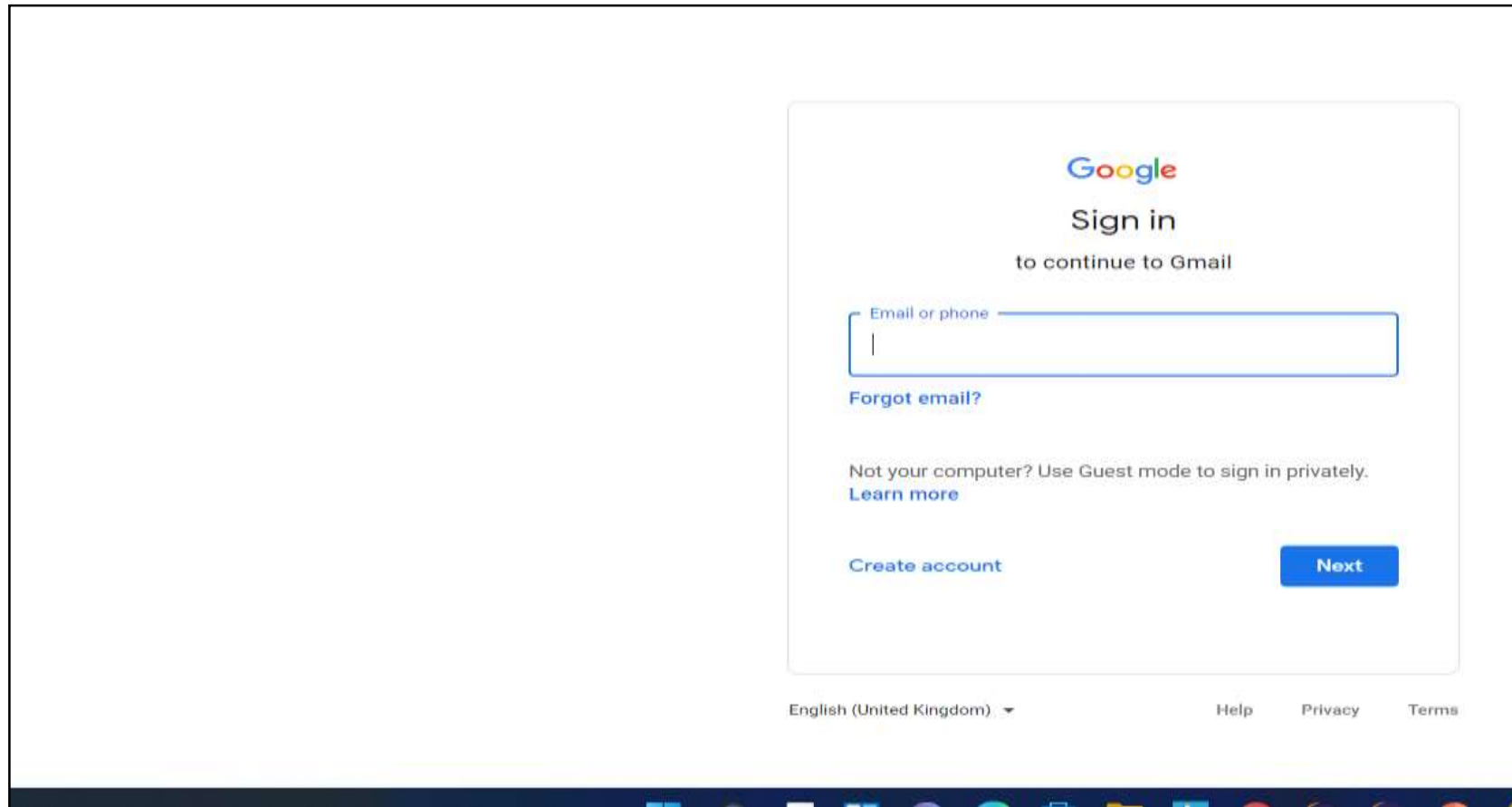
Output



Program 3

```
public class DemoAsync{  
    public static void main(String[] args) {  
WebDriver driver = new ChromeDriver();  
        long et;  
        driver.get("https://www.gmail.com/index.htm");  
        long ct = System.currentTimeMillis();  
        JavascriptExecutor js = (JavascriptExecutor) driver;  
        js.executeAsyncScript("window.setTimeout(arguments[arguments.length - 1], 500);");  
        et=System.currentTimeMillis() - ct;  
        System.out.println("Difference in time"+et);  
        driver.quit();  
    }  
}
```


Output



The image shows a screenshot of the Google Sign-in page for Gmail. The page has a white background with a light gray border. In the center, there is a white box containing the Google logo, the text "Sign in to continue to Gmail", a text input field labeled "Email or phone", a link for "Forgot email?", a link for "Not your computer? Use Guest mode to sign in privately. Learn more", a link for "Create account", and a blue "Next" button. At the bottom of the page, there is a footer with the text "English (United Kingdom)" and a dropdown arrow, followed by links for "Help", "Privacy", and "Terms".

Google

Sign in
to continue to Gmail

Email or phone

[Forgot email?](#)

Not your computer? Use Guest mode to sign in privately.
[Learn more](#)

[Create account](#) [Next](#)

English (United Kingdom) ▼ [Help](#) [Privacy](#) [Terms](#)

Quiz



1) Horizontal window scrolling is done by `executescript`.

a) True

b) False

Answer : Option a)

Quiz



2) You can redirect the webpage in JavaScript by

a) redirect

b) relocate

c) navigate

d) location

Answer : Option d)

Quiz



3) Which JavaScript function is used to execute another function after a given time interval.

a) Timeout()

b) TimeInterval()

c) setTimeout

d) All of the above

Answer : Option c)

Quiz



4) How can you get the total number of arguments passed to a function?

a) Using args.length property

b) Using arguments.length property

c) Both

c) None

Answer : Option b)

THANK YOU